# CMPT 225 Assignment 1 Rubric

| Criterion | Full credit<br>complete and correct | Most credit<br>minor flaws only | Partial credit<br>significant gaps | Little or no credit<br>missing or mostly wrong |
|---|---|---|---|---|
| **Correctness**<br>*Applies to: all parts* | Every requested item is correct and clearly stated; nothing required is missing. | Correct in substance, with one small slip that does not change the main result. | Some items are correct, but a significant error or omission affects the result. | Mostly incorrect, off topic, or blank. |
| **Reasoning and work shown**<br>*Applies to: any justification or explanation* | A complete, logical argument; every step supports the answer; length and form match what is asked. | Sound reasoning with one small gap or skipped step that is easy to fill in. | Vague or partly incorrect, or the answer is restated without support. | No supporting work, or the work contradicts the answer. |
| **Definitions, notation and conventions**<br>*Applies to: A, C, D2–D3* | Definitions are precise and complete; symbols and stated conventions are used correctly. | Precise apart from one small omission or an inconsistent symbol. | The idea is recognizable but stated loosely, or conventions are applied inconsistently. | The definition is incorrect, or notation is misused so that the meaning is lost. |
| **Complexity analysis**<br>*Applies to: C1, D2–D3, E2, G1* | Uses the right variable, and is justified as required. | Correct bound, with a small flaw in the supporting argument. | The bound is valid but stated convention is ignored. | The bound is incorrect, unsupported, or missing. |
| **Diagrams**<br>*Applies to: D1(a), E1(a)–(b), E3(a)* | All required nodes and references drawn, correctly directed and labelled; null shown as ∅. | Structure is right apart from one missing or misdirected arrow, or a missing label. | Several references are wrong or missing, or the drawing is ambiguous. | The structure is fundamentally wrong, or not drawn as boxes and arrows. |
| **Code**<br>*Applies to: D3(a), E2, E3(c), G1(b)* | Correct in general and for each special case named; meets every stated constraint. | Correct overall, with one minor defect in a special case. | A reasonable idea, but it fails on some inputs or breaks a stated constraint. | The approach would not work, is unrelated to the question, or is missing. |
| **Tracing and debugging**<br>*Applies to: D1, D3(c), E3* | State is tracked accurately, step by step; a fault's cause and effect are explained. | One step is mis-tracked, but the rest of the trace is sound. | Tracing is partly correct, or a fault's symptoms are described without its cause. | State is not tracked, or the conclusions do not match the code given. |
| **Clarity and presentation**<br>*Applies to: all parts* | Answers are labelled by question number, legible and concise; code is indented and readable. | Occasional lapses that do not hinder reading. | Hard to follow in places; the marker must guess at your intent. | Illegible or disorganized, so credit cannot be assigned. |

### How marks are awarded

- Where a question splits its marks into labelled components (for example, a verdict and its justification, or running time and space), each component is assessed separately against the criteria above.
- A correct final answer without the requested justification or work earns only part of the marks.
- Code may be written in Java or in clear Java-like pseudocode. Use only the library classes a question explicitly allows.

**Before you submit:** ☐ every sub-part answered and labelled ☐ work shown where requested ☐ stated conventions followed ☐ special cases checked in your code ☐ name and student ID filled in